# Tony Siu

Founding AI Engineer · Multi Modal GenAI researcher · Applied Statistics Researcher

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## Professional Experience

### Agent Engineer III

*ZoomInfo* — *Apr 2026 - Present*

- Reduced agent team execution costs by **over 90%**, from **tens of thousands of dollars to hundreds per run**, by engineering a GCP code execution sandbox with GCS data offloading.
- Enabled AI agents to analyze **200,000-row datasets** and **gigabytes of data**, offloading **millions of tokens** beyond LLM context limits.
- Built and hardened MCP tooling for ZoomInfo's Universal Agent API, unifying access to **3 agent categories—MCP agents, Agent Teams, and conversational intelligence**—with registry reconciliation, tenant isolation, and actionable execution diagnostics.
- Expanded audience analysis across **3 integration surfaces—public MCP tools, Agent Teams, and workbench processing**—with validated artifact caching and context isolation to prevent cross-run data contamination.
- Developed **4 evaluation datasets** grounded in product use cases and identified **3 platform bugs, including 2 critical issues**, strengthening AI agent reliability and production readiness.

### Founding AI Engineer — *San Francisco, CA*

*A16z Speedrun SR005* — *Apr 2025 - Dec 2025*

- Architected & built a Natural-language-to-full GenAI game generation MVP for games (engine, assets, logic) in under 30 days
- Principal Engineer that delivered ~60% of core engineering output of a 20-person startup, while owning architecture, execution, and delivery
- 200× performance optimization to reduce sprite/GIF rendering from 5s per animation to 0.2s across 78+ concurrent assets
- Developed vision-based, promptless agentic loops for long-context, cross-session user memory and task automation
- Reimplemented bleeding-edge research (ACE, Tongyi, AgentFold papers) into production-grade CUA scaffolds and production agent runtimes

### Code&Coffee Philadelphia Chapter Founder — *Philadelphia, PA*

*Code&Coffee* — *Jan 2024 - Present*

- Founded and led Philadelphia's largest active, builder-focused nonprofit tech community, scaling from 2 founders to 3,500+ members in one year
- Built a cross-domain ecosystem connecting engineers, founders, researchers, VCs, and mentors across AI, systems, and product
- Produced high-impact programming including Philly Tech Week 2025's only builder-focused hackathon, founder demo days, and technical meetups (AI papers, coworking, LeetCode, coffee-and-code)
- How Tony Siu built Philly's 3k+ member Code and Coffee meetup. **Technical.ly**, 2026.
- Code & Coffee Philadelphia's Mission to Revive Community-Driven Learning. **TechTimes**, 2026.
- How Research Projects & Community Work Shape His Approach to Building. **IBTimes**, 2026.

### Multi-Modal Generative AI Researcher — *Philadelphia, PA*

*Temple College of Science & Technology* — *May 2023 - May 2025*

- Authored multiple papers submitted to IEEE, CAIP, and ECAI on multimodal computer vision and video understanding
- Improved performance on video QA performance by 6% using rPPG features from RhythmFormer and RhythmMamba over baseline fine-tuning
- Achieved 6% in- and out-of-sample generalization gains through Direct Preference Optimization without RLHF
- Delivered up to 9% performance improvements across benchmarks using preference alignment methods (KTO, IPO, RSO, cDPO, SSPO, GRPO)
- Generated synthetic preferred/rejected response pairs to enable scalable preference optimization without human annotation
- Evaluated SOTA video–language models (MiniGPT-4-Video, LLaVA-Next, VideoMamba) across DAiSEE, EngageNet, and SED datasets
- Reached 97% F1 on student engagement detection via engagement-level prompt fine-tuning for visual question answering

### Google UR Computer Vision Research Lead — *Taipei, Taiwan*

*Google Research* — *Jan 2022 - May 2023*

- Led computer vision research for improving Google Pixel automated UI defect detection pipelines
- Secured continued Google Level-2 sponsorship through novel multimodal computer vision research outcomes
- Boosted UI flicker detection from 43% to 98% precision–recall performance using a CNN–Transformer hybrid architecture
- Expanded training data 10× via targeted augmentation, scaling from 180 to 1,991 labeled video snippets
- Reduced model training time by 20× through informed weight initialization, cutting convergence from 1,000 to 50 epochs

### Data Analytics Intern — *Philadelphia, PA*

*Symetri USA* — *Jun 2024 - Sept 2024*

- Abstracted away traditional data infrastructure using AIaaS paradigm, maintaining client data security and minimal technological overhead
- Migrated data infrastructure to Multi-Agent GenAI BIM Vector and Graph based Retrieval Augmented Generation POCs for client demonstration
- Landed partnerships with PennMedicine, several other Hospital Authorities and Port Authorities in PANJNY through built Compliance Frontend

### AI Specialist Engineer — *Taipei, Taiwan*

*Taiwan Tech Arena* — *May 2020 - Oct 2020*

- Achieved real-time 360° video stitching on NVIDIA Jetson Xavier, reducing dual 4K fisheye (170°) composePanorama latency from 6s to 0.2s
- Optimized OpenCV stitching pipeline by CUDA-integrating OpenCV 2.4 image blending into OpenCV 4.5, reducing frame time from 0.2s to 0.01s
- Enabled real-time stitching via single-pass EstimateTransform, multithreading, and stationary-camera assumptions for continuous 360° capture
- Delivered 6× faster OpenCV CUDA/cuDNN builds by packaging a single optimized compilation as reusable CPack Debian binaries
- Implemented real-time 3D-to-2D homography projections to map player field positions from panoramic video feeds
- Developed real-time sports video analytics including camera panning, zoom detection, and slow-motion event analysis

## Education

**Masters, Computational Data Science** | GPA 4.0
*Temple University*, Philadelphia, PA
Jan 2024 - Current

**Bachelors, Computational Data Science**
*Temple University*, Philadelphia, PA
Aug 2022 - Dec 2024

## Publications

Generalizable Detection of Student Engagement in Online Learning Environments. CAIP 2025.

rPPG Signal Enhances Student Engagement Recognition in Online Video Education. *IEEE*, 2025.